

Research on Ideological and Political Education Effect Evaluation Model Based on Multimodal Data Fusion and Attention Mechanism

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Abstract

In response to the core issues of strong subjectivity and single dimensionality in traditional evaluation methods for ideological and political education, this paper proposes an evaluation model based on multimodal data fusion and an attention mechanism, named the Multimodal Attention-based Ideological and Political Education Evaluation (MA-IEE). This model aims to achieve an objective and quantitative assessment of educational effectiveness by intelligently analyzing students' multimodal behavioral data (text, speech, vision) within an educational context, integrating deep learning with a cross-modal attention mechanism. In practical teaching applications, this model can serve as a formative assessment tool to provide teachers with immediate, fine-grained feedback on students' classroom reactions (such as emotional states) and learning engagement. This helps educators dynamically optimize teaching strategies, thereby connecting abstract educational goals with observable and analyzable student behavioral data, rather than being used merely for summative judgments. To verify the technical feasibility of the model, extensive comparative experiments and ablation studies were conducted on the publicly available multimodal emotion analysis datasets, CMU-MOSEI and IEMOCAP. The results indicate that the proposed MA-IEE model outperforms models based on single modalities and traditional fusion methods across key metrics such as accuracy, precision, recall, and F1-score, thus validating its effectiveness and superiority in fusing multimodal behavioral data for quantitative evaluation.

CCS Concepts

• **Theory of computation** → Theory and algorithms for application domains; Machine learning theory.

Keywords

Ideological and Political Education, Effect Evaluation, Multimodal Data Fusion, Attention Mechanism, Deep Learning

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1 Introduction

As a key course for implementing the fundamental task of "fostering virtue through education," the scientific evaluation of the effectiveness of Ideological and Political Education (IPE) is a core component for improving educational quality and optimizing teaching strategies [1]. For a long time, the evaluation of IPE effectiveness has primarily relied on traditional methods such as questionnaires, interviews, and course essays. Although these methods can reflect students' cognition and attitudes to some extent, their inherent limitations have become increasingly prominent [2]. On one hand, evaluation results are susceptible to interference from students' subjective intentions and social expectations, making their authenticity and objectivity difficult to guarantee. On the other hand, the evaluation dimensions are relatively singular, often focusing on students' knowledge acquisition and theoretical identification, which makes it difficult to comprehensively and dynamically capture the deep-level changes in students' emotions, attitudes, and behaviors [3].

With the rapid development of artificial intelligence and big data technologies, it has become possible to conduct fine-grained, intelligent analysis of the educational process. Multimodal learning, as an important branch of artificial intelligence, aims to achieve more accurate decision-making and prediction than single-modality approaches by enabling machines to understand and fuse data from multiple information sources (such as text, images, and speech) [4]. In educational settings, a student's learning state and cognitive engagement are not determined by a single behavior but are comprehensively reflected through a combination of multiple modalities, including the textual content of classroom speech, the emotional tone of their voice, and the level of concentration shown in their facial expressions [5]. Therefore, the introduction of multimodal data fusion technology can characterize students' learning status from richer dimensions, providing a new paradigm for the reform of IPE effectiveness evaluation.

However, a simple concatenation or linear superposition of multimodal data often overlooks the complex interactions and information redundancy between different modalities, making it difficult to achieve effective fusion. Inspired by human visual attention, the core idea of the Attention Mechanism is to enable a model to "focus" on the most critical information for the current task while ignoring secondary information [6]. Introducing an attention mechanism into the multimodal fusion process allows the model

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to adaptively learn the importance of different modalities in a specific evaluation task, as well as the feature correlations within and between modalities, thereby achieving more efficient and precise information integration [7].

Based on the above analysis, this paper aims to explore an intelligent IPE evaluation pathway that integrates multimodal data and an attention mechanism. To this end, an evaluation model named MA-IEE is constructed. This model can synchronously process students' text, audio, and video data during the learning process and effectively fuse information through a cross-modal attention module to ultimately output a quantitative evaluation grade. This study hopes that this data-driven evaluation paradigm can overcome the drawbacks of traditional methods and provide more objective, scientific, and fine-grained decision support for the quality monitoring and teaching improvement of ideological and political education.

2 Related Theories

2.1 Multimodal Data Fusion

Multimodal Data Fusion aims to integrate data from different modalities to generate a more comprehensive and accurate representation. Depending on the level of abstraction at which fusion occurs, it is generally divided into Early Fusion, Late Fusion, and Hybrid Fusion. Early fusion is performed at the data or feature level, directly concatenating the feature vectors of each modality. While simple to implement, it may yield suboptimal results due to feature dimension mismatch and information redundancy [8]. Late fusion occurs at the decision level, where each unimodal model makes an independent prediction, and the results are then combined through voting or weighting. This method offers high flexibility but ignores early-stage interactions between modalities. Hybrid fusion combines the first two, attempting to find a balance between the feature and decision levels to capture complex inter-modal correlations, and it is a current research hotspot.

2.2 Attention Mechanism

The attention mechanism first achieved great success in machine translation tasks within natural language processing and has since been widely applied to fields such as computer vision and speech recognition. Its core is a weighted summation process, where weights are obtained by calculating the similarity between a Query and a set of Keys. These weights are then used to perform a weighted sum of the corresponding Values. Self-Attention focuses on the dependencies between different parts within the same modality, while Cross-Modal Attention is used to compute the correlation between features of different modalities, allowing one modality to "guide" the feature representation of another [9]. This makes it highly suitable for multimodal fusion tasks.

2.3 Theories of Ideological and Political Education Effect Evaluation

Traditional theories for evaluating the effectiveness of ideological and political education primarily revolve around Kirkpatrick's Four-Level Training Evaluation Model, which includes Reaction, Learning, Behavior, and Results levels. However, acquiring data for these levels is difficult and highly subjective. With the advancement

of educational informatization, Learning Analytics technology has been introduced into the field of educational evaluation [10]. It emphasizes understanding and optimizing the learning process by collecting and analyzing the vast amounts of data generated by learners. This study is based on the philosophy of learning analytics, treating students' multimodal classroom behavior data as the object of analysis. It attempts to objectively quantify the states at the Learning and Behavior levels, thereby achieving an in-depth evaluation of educational effectiveness.

3 Model Construction

To achieve a precise evaluation of the effectiveness of ideological and political education, this paper designs an evaluation model based on multimodal data fusion and an attention mechanism (MA-IEE). The model is primarily composed of three parts: a unimodal feature extraction module, a cross-modal attention fusion module, and an evaluation result prediction module.

3.1 Unimodal Feature Extraction Module

This module is responsible for transforming the raw text, visual, and audio data streams into fixed-dimension feature vectors, which serve as the input for the subsequent fusion module.

Textual Feature Extraction: For textual data, a pre-trained BERT model is employed. An input text sequence $T = \{w_1, w_2, \dots, w_n\}$ is processed into a format compatible with BERT's input. The hidden state vector corresponding to the token from BERT's output is taken as the aggregated semantic representation of the entire sequence.

$$F_t = \text{BERT}(T)_{[CLS]} \in \mathbb{R}^{d_t} \quad (1)$$

where d_t is the dimension of the textual feature.

Temporal Feature Extraction for Vision/Audio: For the temporal modalities of vision and audio, this paper uses a Long Short-Term Memory (LSTM) network to capture their dynamic information. LSTM effectively addresses the long-sequence dependency problem through its gating mechanisms (input gate, forget gate, output gate). At time step k , the computation process of an LSTM unit is as follows:

$$i_k = \sigma(W_i x_k + U_i h_{k-1} + b_i) \quad (2)$$

$$f_k = \sigma(W_f x_k + U_f h_{k-1} + b_f) \quad (3)$$

$$o_k = \sigma(W_o x_k + U_o h_{k-1} + b_o) \quad (4)$$

$$c_k = f_k \odot c_{k-1} + i_k \odot \tanh(W_c x_k + U_c h_{k-1} + b_c) \quad (5)$$

$$h_k = o_k \odot \tanh(c_k) \quad (6)$$

where x_k is the input feature at the current time step (frame-level features extracted by ResNet for the visual modality, and MFCCs for the audio modality), while h_{k-1} and c_{k-1} are the hidden state and cell state from the previous time step, respectively. i_k, f_k, o_k are the activation values of the input, forget, and output gates. W, U, b are learnable parameter matrices and biases, σ is the Sigmoid function, and $\odot$ represents element-wise multiplication. The hidden state at the last time step, h_L is taken as the final feature representation for that modality:

$$F_v = \text{LSTM}_v(X_v) \in \mathbb{R}^{d_v}, F_a = \text{LSTM}_a(X_a) \in \mathbb{R}^{d_a} \quad (7)$$

where X_v, X_a are the visual and audio feature sequences, and d_v, d_a are the corresponding feature dimensions.

3.2 Cross-Modal Attention Fusion Module

This module is the core of the model, responsible for the effective fusion of features extracted from different modalities. This paper designs a cross-modal attention mechanism guided by the textual modality. This design is based on the hypothesis that in cognitively demanding scenarios like ideological and political education, textual content most directly reflects students' cognition and viewpoints. Using it as a guide can more effectively integrate auxiliary behavioral information such as facial expressions and tone of voice.

The mechanism follows the standard Scaled Dot-Product Attention paradigm. Taking the guidance of visual feature F_v by textual feature F_t as an example, F_t is first mapped to a query vector Q , and F_v is mapped to a key vector K and a value vector V :

$$Q = F_t W_Q, K = F_v W_K, V = F_v W_V \quad (8)$$

where $W_Q \in \mathbb{R}^{d_t \times d_k}$, $W_K \in \mathbb{R}^{d_v \times d_k}$, $W_V \in \mathbb{R}^{d_v \times d_{val}}$ are learnable linear transformation matrices, and d_k and d_{val} are the dimensions of the keys and values, respectively.

Subsequently, attention scores are obtained by calculating the dot product of Q and K . After scaling and Softmax normalization, attention weights are derived and used to perform a weighted sum on V , yielding the fused visual feature F'_v :

$$F'_v = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (9)$$

where $\sqrt{d_k}$ is the scaling factor used to prevent vanishing gradients.

Similarly, the text-guided audio feature F'_a can be obtained. Finally, the original textual feature is concatenated with the attention-weighted visual and audio features to form the final fused feature vector F_{fused} :

$$F_{fused} = \text{Concat}(F_t, F'_v, F'_a) \quad (10)$$

Compared to other more complex fusion strategies (such as tensor fusion), the cross-modal attention mechanism adopted in this paper ensures fusion effectiveness while offering better interpretability and lower computational complexity, making it more suitable for deployment and analysis in educational settings.

3.3 Evaluation Result Prediction and Loss Function

The fused feature vector F_{fused} is fed into a Multilayer Perceptron (MLP) for classification.

$$P = \text{softmax}\left(W_2 \left(\text{ReLU}\left(W_1 F_{fused} + b_1\right)\right) + b_2\right) \quad (11)$$

where W_1 , b_1 , W_2 , b_2 are the weights and biases of the MLP.

The model is optimized using the Cross-Entropy Loss function, which is defined as follows:

$$L_{CE} = - \sum_{i=1}^C y_i \log(p_i) \quad (12)$$

where C is the total number of evaluation grade categories, y_i is a one-hot vector where the value is 1 if the sample's true class is i and 0 otherwise, and p_i is the model's predicted probability that the sample belongs to class i . The training objective is to minimize this loss function.

4 Experimental Verification

4.1 Datasets and Proxy Task

Although the application of multimodal learning in the educational field is increasingly widespread, there is still a scarcity of publicly available, large-scale benchmark datasets specifically for evaluating the effectiveness of ideological and political education. Therefore, to verify the technical feasibility and effectiveness of the proposed model in fusing multimodal behavioral data, this study uses two internationally recognized multimodal emotion analysis datasets, CMU-MOSEI and IEMOCAP, as experimental platforms and defines a Proxy Task.

The rationale for choosing emotion analysis as a proxy task is as follows: While emotional state is not completely equivalent to the ultimate outcome of ideological and political education (such as value internalization), it is a critical and observable intermediate link in the evaluation process. According to the Kirkpatrick evaluation model, students' classroom reactions (Reaction level) and learning engagement (Learning level) are important components of the assessment. Students' emotional states in the classroom (such as positive, focused, confused, or resistant) are direct manifestations of their reactions and engagement. This setup is also supported by the control-value theory of achievement emotions in educational psychology, which posits that students' positive emotions (e.g., enjoyment, hope) are significantly and positively correlated with their learning engagement and the mobilization of cognitive resources. Therefore, this study uses emotion recognition as a proxy task to simulate the assessment of students' classroom participation status. Success in this proxy task can demonstrate that the proposed model has the technical capability to analyze complex multimodal human behavior, laying the groundwork for its future application in real, more complex IPE scenarios.

4.1.1 Dataset Format Description. To ensure the reproducibility of the experiments, the format of the datasets used is described here. Each sample in the datasets consists of data from three modalities and its corresponding label.

CMU-MOSEI: Each sample in this dataset is an independent video segment. Its data structure mainly includes:

- **Text:** Manually transcribed speech content, in string format.
- **Vision:** The original sequence of video frames. In this study, features from each frame are extracted using a ResNet network to form a sequence of feature vectors.
- **Audio:** The original audio file in WAV format. In this study, MFCCs acoustic features are extracted from it.
- **Label:** A continuous floating-point number between $[-3, 3]$, representing emotional valence.

IEMOCAP: This dataset consists of multiple sessions, each of which is segmented into several utterances, with each utterance being a sample. Its data structure mainly includes:

- **Text:** Manually transcribed text corresponding to the utterance, in string format.
- **Vision:** A sequence of video frames recording the actor's facial expressions and movements. Processed in the same manner as above.
- **Audio:** The WAV format audio file corresponding to the utterance. Processed in the same manner as above.

- **Label:** A discrete emotional category string, such as 'happy', 'sad', etc.

4.1.2 Dataset Partition. The data was processed according to the format described above and partitioned as per official recommendations. For the CMU-MOSEI dataset, the samples were divided into a training set (16,265), a validation set (1,871), and a test set (4,643). For the IEMOCAP dataset, the samples were divided into a training set (4,425), a validation set (1,106), and a test set (2,229). The number of classification categories for both tasks was set to 4.

4.2 Experimental Settings

All experiments were conducted using the PyTorch framework. The AdamW optimizer was used with a learning rate set to 1×10^{-5} , a batch size of 32, and 50 training epochs. An early stopping strategy was employed to prevent overfitting. The feature dimensions were set as follows: $d_t=768$, $d_v=512$, $d_a=128$. In the attention module, both d_k and d_{val} were set to 128.

4.3 Main Results and Analysis

4.3.1 Evaluation Metrics. To quantitatively evaluate the model's performance, this study employs the following four standard metrics based on the Confusion Matrix. For a multi-class classification problem, the following four basic concepts are first defined:

- **True Positive (TP):** The actual class of the sample is positive, and the model also predicts it as positive.
- **False Positive (FP):** The actual class of the sample is negative, but the model predicts it as positive.
- **True Negative (TN):** The actual class of the sample is negative, and the model also predicts it as negative.
- **False Negative (FN):** The actual class of the sample is positive, but the model predicts it as negative.

Based on these definitions, the evaluation metrics are calculated as follows:

Accuracy: Represents the proportion of correctly classified samples to the total number of samples. It is the most intuitive metric for measuring the overall performance of the model.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (13)$$

Precision: Refers to the proportion of actual positive samples among all samples predicted as positive by the model. It measures the accuracy of the predictions.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (14)$$

Recall: Refers to the proportion of correctly predicted positive samples among all actual positive samples. It measures the model's ability to find all positive instances.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (15)$$

F1-Score: The harmonic mean of Precision and Recall, used for a comprehensive evaluation of the model's performance, especially reliable in cases of class imbalance.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

As this study involves a multi-class classification task, all metrics except for accuracy are calculated using a weighted average to eliminate the impact of class imbalance.

4.3.2 Experimental Results. To validate the effectiveness of the proposed MA-IEE model, the following baseline models were set up for comparison:

- **Text-BERT:** Uses only the text modality.
- **Video-LSTM:** Uses only the visual modality.
- **Audio-LSTM:** Uses only the audio modality.
- **Early Fusion:** Directly concatenates the feature vectors of the three modalities.
- **MA-IEE (Proposed Model):** The model employing the cross-modal attention fusion mechanism proposed in this paper.

The following tables, Table 1 and Table 2, respectively, show the final performance of all comparative models on the CMU-MOSEI and IEMOCAP test sets. The data in the tables are the results calculated according to the evaluation metrics defined in section 4.3.1. Higher values indicate better model performance on that specific metric. By comparing these data, the effectiveness of the proposed MA-IEE model relative to other methods can be clearly assessed.

As can be seen from the data analysis in Table 1 and Table 2, the proposed MA-IEE model achieved the best performance on all evaluation metrics in both monologue scenarios (CMU-MOSEI) and dialogue scenarios (IEMOCAP), demonstrating the model's effectiveness and a certain degree of generalization ability. Compared to the second-best performing baseline model (Early Fusion), the MA-IEE model shows a significant advantage. Specifically, using the comprehensive F1-score as a benchmark, on the CMU-MOSEI dataset, our model (0.833) improved by 3.3 percentage points over the Early Fusion model (0.800). On the more complex IEMOCAP dataset, our model (0.681) improved by 3.8 percentage points over the Early Fusion model (0.643). This result not only highlights the significant advantage of the cross-modal attention mechanism in intelligently selecting and fusing information but also reveals an interesting phenomenon: our model achieved more significant performance gains in the more interactive dialogue scenario. This may indicate that in scenarios with more dynamic changes, our model's attention mechanism can more effectively capture and utilize the instantaneous correlations between modalities, which has positive implications for analyzing real classroom discussion scenarios.

4.4 Ablation Study

To further investigate the contribution of each component of the model, a series of ablation experiments were conducted on the CMU-MOSEI dataset. Table 3 shows the changes in model performance after removing different modalities or changing the attention guidance method. All results were obtained on the CMU-MOSEI test set. By analyzing the changes in these data, the contribution of each module to the overall performance of the model can be quantified.

The results of the ablation study clearly reveal:

Table 1: Performance comparison of different models on the CMU-MOSEI test set

Model	Accuracy	Precision	Recall	F1-Score
Text-BERT	0.785	0.783	0.785	0.781
Video-LSTM	0.512	0.509	0.512	0.507
Audio-LSTM	0.563	0.558	0.563	0.559
Early Fusion	0.801	0.799	0.801	0.800
MA-IEE (Proposed)	0.834	0.832	0.834	0.833

Table 2: Performance comparison of different models on the IEMOCAP test set

Model	Accuracy	Precision	Recall	F1-Score
Text-BERT	0.621	0.619	0.621	0.620
Video-LSTM	0.455	0.451	0.455	0.452
Audio-LSTM	0.583	0.580	0.583	0.581
Early Fusion	0.645	0.642	0.645	0.643
MA-IEE (Proposed)	0.682	0.680	0.682	0.681

Table 3: Ablation Study Results (CMU-MOSEI)

Model Variant	Accuracy	Precision	Recall	F1-Score
MA-IEE (Full Model)	0.834	0.832	0.834	0.833
w/o Text (no text mode)	0.615	0.611	0.615	0.612
w/o Vision (no visual modality)	0.820	0.818	0.820	0.819
w/o Audio (no audio mode)	0.812	0.810	0.812	0.811
Vision Query (visual guidance)	0.815	0.813	0.815	0.814

- After removing the text modality, the model’s performance dropped precipitously, proving the central role of textual content in the evaluation task.
- Removing the visual or audio modality also led to a slight decrease in performance, indicating that these two modalities provide beneficial auxiliary information and verifying the necessity of multimodal fusion.
- Changing the guiding modality from text to vision resulted in a performance drop of nearly 2 percentage points, which strongly supports the rationality of this paper’s design philosophy of “using textual content as the core guide to integrate auxiliary behavioral information.”

4.5 Limitations and Future Outlook

This study also has certain limitations. First, as previously mentioned, emotional state does not fully represent the entire effect of ideological and political education, which also includes deeper levels of cognitive construction and value identification. The conclusions of this study are based on a proxy task, and its effectiveness in real IPE classroom scenarios still requires further validation. Second, the design of the model in this paper is geared towards generality and has not been deeply customized for the specific discourse systems and behavioral patterns (such as debate, contemplation, etc.) of ideological and political education.

Future research can be expanded in the following aspects: First, efforts should be made to construct a multimodal dataset oriented

towards real IPE scenarios, containing fine-grained annotations. Second, domain knowledge could be integrated into the model, for example, by designing modules capable of recognizing specific IPE terminology and dialectical logic to achieve more precise evaluation.

4.6 Ethical Considerations

The application of artificial intelligence technology to analyze and evaluate students’ classroom behavior must attach great importance to related ethical issues. The first is student privacy and data security. The collection of students’ audio, video, and text data constitutes sensitive information and must be carried out only after obtaining explicit informed consent from the students. All data should undergo strict anonymization and encryption during storage, transmission, and processing to prevent data leakage and misuse. Second is the fairness and purpose of the evaluation. Such technology should be used as a formative assessment tool to help teachers improve teaching and understand student situations, rather than as a summative evaluation basis for ranking or punishing students. It is necessary to be vigilant against the potential “labeling” effect of technology and avoid reducing complex individuals to cold evaluation scores. Finally, there is the issue of model interpretability and transparency. In future practical applications, developing explainable AI (XAI) models is crucial. When the system gives an evaluation result, it should be able to provide the teacher with the main basis for that judgment (e.g., “The student’s positive facial expression and affirmative tone of voice provided strong support when

discussing key concepts”). This not only enhances the teacher’s trust in the system but also aligns with the ethical requirement that educational evaluation should be people-oriented, transparent, and fair.

5 Conclusion

In response to the problems of insufficient objectivity and single dimensionality in traditional evaluation methods for ideological and political education, this paper has explored and proposed an intelligent evaluation model based on multimodal data fusion and an attention mechanism (MA-IEE). The model extracts features from text, visual, and audio data through parallel deep learning networks and innovatively designs a cross-modal attention module to dynamically and selectively fuse multi-source information. Through extensive experiments on two public datasets, CMU-MOSEI and IEMOCAP, including comparisons with baseline models and in-depth ablation studies, the results show that the proposed model is technically capable of effectively fusing multimodal behavioral data, and its performance demonstrates significant superiority over unimodal models and simple fusion methods. Despite certain limitations, this research proves that the model can more deeply understand and quantify students’ comprehensive performance in the educational process, providing a feasible technical path and valuable reference for achieving objective, comprehensive, and efficient evaluation of ideological and political education in the future. This study not only offers a new paradigm for IPE evaluation but also

provides a beneficial interdisciplinary exploration for the deep and responsible application of artificial intelligence technology in the humanities and social sciences.

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